COS790 Project 5: Final Paper

Using Explainable AI to Understand and Improve Cross-Domain Selection Perturbative Hyper-Heuristics

Liam van Kasterop u22539761
Dreas Vermaak u22497618, Katleho Moeletsi 19061430,
Emmanuel Ramphabana u25721373

Department of Computer Science, University of Pretoria

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1 Introduction

The Project 5 brief asks for explainable artificial intelligence (XAI) to be used to *understand and improve* the performance of selection hyper-heuristics, examining both single-point and multi-point search on the cross-domain CHeSC benchmark. Assessments 3 to 5 built the measurement apparatus for this: four selection perturbative hyper-heuristics run on the six real HyFlex domains, a trace of the search state recorded before every heuristic application, and SHAP and LIME surrogates checked for fidelity and stability across ten seeds.

The feedback on Assessment 5 was that this delivered only half of the project. SHAP and LIME were implemented correctly, but they were never used to improve the hyper-heuristics, and the research questions did not lead there. Looking back, the reason is structural. The Assessment 5 surrogates predicted whether a candidate was *accepted*, so they described the move acceptance criterion rather than the selection decision. They also could not see which heuristic was being applied: the `feat_last_h*` features record the previous heuristic, and the current one was not a feature. An explanation of that model cannot say which heuristic should be chosen, so it cannot suggest how to choose better.

This paper closes that gap. The XAI layer now explains heuristic *effectiveness*: which heuristic types produce improving moves, in which search states and phases. A fixed procedure turns those explanations into a change to the hyper-heuristic. The changed hyper-heuristic is tested on held-out data, and the explanations are then recomputed to check that it behaves as they predicted. Figure 1 shows the loop.

The overarching research question of the project is the aim set by the brief:

> **Overarching question.** Can explainable AI be used to understand and improve the performance of single-point and multi-point selection hyper-heuristics across the domains of the CHeSC cross-domain challenge?

Four research questions break this down, one for each step of the loop:

- **RQ1 (Understand).** Which heuristic types, in which search states and phases, produce improving moves in each domain?
- **RQ2 (Improve).** Do changes derived from the RQ1 explanations significantly improve SP-CF-AILTA and MP-CF over their unmodified versions?
- **RQ3 (Generalise).** Do the changes still help on a domain whose traces were not used to derive them?

- **RQ4 (Mechanism).** Did the improved variants change behaviour in the way the explanations predicted, and do they beat equally sized changes that were not chosen by XAI?

RQ1 covers the "understand" part of the overarching question, RQ2 the "improve" part, RQ3 the cross-domain part, and RQ4 tests whether any improvement is due to the explanations rather than to chance or to any restriction of the heuristic set. The performance comparison against Simple Random, which formed RQ1 and RQ2 in Assessment 5, now serves as the baseline the loop starts from. Section 5 gives a direct answer to each research question and then to the overarching question.

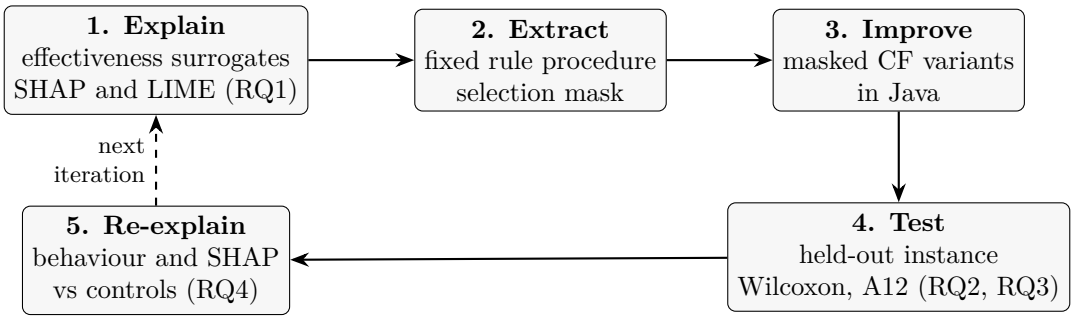


Figure 1: The explain, improve, test and re-explain loop used in this study. Assessment 5 stopped after the first box, and its surrogates explained acceptance rather than effectiveness.

2 Hyper-Heuristics Under Study

All four are selection perturbative hyper-heuristics in the sense of Pillay and Qu (2018), and they are unchanged from Assessment 5 and are listed in Table 1. The Modified Choice Function (Drake, Özcan and Burke, 2012) scores each heuristic h as $\mathrm{CF}(h) = \phi f_1[h] + \phi f_2[prev][h] + (1-\phi) f_3[h]$, where f_1 is the decayed improvement per unit CPU time of h, f_2 is the pair-sequence synergy with the previous heuristic, and f_3 is the time since h was last applied. The weight ϕ is reset to 0.99 on an improving move and decays by 0.01 per non-improving step, so the f_3 term, which favours idle heuristics, takes over as the search stagnates. Late Acceptance (Burke and Bykov, 2017) accepts a candidate that is no worse than the incumbent of L iterations earlier. AILTA is a linear-decay variant of the threshold acceptance used in AdapHH (Mısır et al., 2012), with threshold $\theta(t) = 0.02\,|f_0|\,(1 - t/T)$ and no reheating. MP-CF keeps a population of ten, replaces the binary-tournament worst individual only with a strictly better candidate (never the population best), and shares one Choice Function across the population.

Technique	Topology	Selection	Acceptance
SP-SR-LA	Single-point	Simple Random	Late Acceptance, $L = 50$
SP-CF-LA	Single-point	Choice Function	Late Acceptance, $L = 50$
SP-CF-AILTA	Single-point	Choice Function	Linear-decay AILTA
MP-CF	Multi-point (10)	Choice Function	Tournament replacement

Table 1: The four base techniques. SP-CF-AILTA and MP-CF are the targets for improvement because they were the strongest single-point and multi-point techniques in Assessment 5.

All runs use the real CHeSC HyFlex domains (Ochoa et al., 2012). Budgets are counted in heuristic applications and are the per-domain budgets set from convergence curves in Assessment 5: SAT 50,000 (11 heuristics), BinPacking 100,000 (8), FlowShop 100,000 (15), TSP 10,000 (13), VRP 50,000 (10) and PersonnelScheduling 5,000 (12). Each heuristic belongs to one HyFlex type class: mutation, ruin-and-recreate, local search or crossover.

3 Method: The XAI Improvement Loop

3.1 Explaining effectiveness instead of acceptance

Each trace row is one heuristic application. The new target label is $improved = 1$ if the objective after the application is strictly lower than before it (all domains are minimisation problems). A second label, $\delta_{null} = 1$ when the objective does not change, is used later to measure wasted applications.

The features are the pre-application state features from Assessment 5, with `feat_llh_count` still excluded because it acts as a domain label, plus the heuristic applied in this row, encoded as a one-hot type class (`h_is_MUTATION`, `h_is_RUIN_RECREATE`, `h_is_LOCAL_SEARCH`, `h_is_CROSSOVER`) and its within-domain index. Including the current heuristic is legitimate here because the question is now "is applying h in state s effective?". All features are computed before the application, and normalisation uses running ranges rather than full-run statistics, so no information from later in the run leaks into a decision-time feature.

3.2 An unconfounded dataset from the Simple Random baseline

Under the Choice Function, which heuristic is applied depends on the search state, so the effect of a heuristic is mixed up with the states in which the Choice Function tends to pick it. Under Simple Random selection the heuristic is drawn independently of the state. The SP-SR-LA traces are therefore a randomised sample of heuristic effectiveness across states, and they are the primary dataset for rule extraction. This turns the Assessment 5 finding that SP-SR-LA acceptance could not be explained into a strength: the baseline that could not be explained as a policy is the cleanest evidence of what each heuristic does. Secondary surrogates are trained on the SP-CF-AILTA and MP-CF traces to describe the states those techniques actually visit.

3.3 Surrogates and explanations

One gradient-boosting classifier (Friedman, 2001) with 100 trees of depth 4 and inverse-frequency class weights is trained per technique, domain and seed, giving 180 surrogates on the 240 Assessment 5 traces (instance 0). Fidelity is measured by five-fold stratified cross-validation, and a surrogate is explained only if its ROC-AUC is at least 0.6 and its PR-AUC is at least twice the positive base rate. SHAP values come from TreeExplainer (Lundberg and Lee, 2017; Lundberg et al., 2020). For the type-class features the signed SHAP value is averaged only over rows where that type was applied, so a negative value means that applying the type pushes the prediction away from improvement. LIME (Ribeiro et al., 2016) is applied at 50 instances spread over search progress and grouped into ten progress bands, as in Assessment 5. Stability is measured by Kendall's τ between per-seed feature rankings.

The main output of this step is an *effectiveness map* with 720 rows (three techniques, six domains, four type classes, ten progress bands). Each row gives the mean predicted improvement probability, the mean signed SHAP contribution of the type, the observed improvement rate in the raw traces as a model-free check, the application count, and the fraction of seeds that agree on the sign of the SHAP contribution.

3.4 From explanation to design change

To show that the change comes from the explanations and not from manual tuning, the rules are produced by a fixed, deterministic procedure from the SP-SR-LA effectiveness map. The ten progress bands are merged into three phases: early (progress below 0.3), middle (0.3 to 0.7) and late (above 0.7). **Rule A** excludes type class T in phase P of a domain when all three conditions hold:

1. the signed SHAP contribution of T is negative in at least 8 of the 10 seeds;

2. the observed improvement rate of T is below half the mean rate for that domain and phase;
3. at least one type class remains allowed.

Condition 1 asks the explanation for a consistent negative effect, and condition 2 confirms it directly from the traces, so a type is excluded only when the model and the raw data agree. Rules are stated at the type-class level because heuristic indices refer to different operators in different domains, while type classes are shared, which is what allows the rules to be transferred across domains (RQ3). A second rule using LIME thresholds to trigger AILTA reheating and a diversity response in MP-CF (Rule B) was specified but not implemented in time.

3.5 Variants and controls

Each base technique receives the four variants in Table 2. The two controls are what allow RQ4 to separate the value of the explanation from the effect of simply removing heuristics. RAND removes the same number of type-class and phase cells as XAI but chooses them at random. STAT removes the same number using only descriptive statistics (mean clipped log return per type and phase), with no surrogate.

Variant	Source of the selection mask
XAI	Rule A applied to the target domain's own effectiveness map.
XAI-LODO	Rule A by majority vote over the other five domains only (leave-one-domain-out).
RAND	Same number of excluded cells as XAI, chosen at random (fixed seed 12345).
STAT	Same number of excluded cells, chosen by mean clipped log return only.

Table 2: Variants built on each base technique. The base technique itself is the fifth arm of each comparison.

3.6 Implementation

The mask is a decorator around the existing `SelectionStrategy` interface, so none of the four base techniques is modified. `SelectionMask` reads the rules file for the technique and domain and reports whether a heuristic is allowed at the current budget progress. `MaskedSelectionStrategy` lets the Choice Function score every heuristic as usual, keeping all f_1, f_2, f_3 and ϕ bookkeeping intact, and then picks the highest-scoring *allowed* heuristic. The trace gains two fields, `feat_mask_active` and `feat_allowed_set_size`. Six new unit tests cover the phase boundaries, exclusions, rule parsing and end-to-end masked selection, and all 32 tests pass.

3.7 Experimental protocol

To avoid deriving and testing rules on the same data, all variants are evaluated on **instance 1** of each domain (the rules come from instance 0) with ten new seeds $\{11, 22, \ldots, 99, 111\}$. Each run uses a fresh problem object, fresh selector state and a fresh random number generator. The planned matrix is 2 base techniques $\times$ 5 arms $\times$ 6 domains $\times$ 10 seeds $=$ 600 runs. All 300 single-point runs completed. Of the 300 multi-point runs, the 60 base MP-CF runs and 20 MP-CF-XAI runs completed, and the remaining 220 could not be completed in the time available (Section 3.8).

Within each domain, arms are compared with two-sided Wilcoxon signed-rank tests (Wilcoxon, 1945) paired by seed, with Holm correction (Holm, 1979) within each base technique and domain, and Vargha and Delaney A_{12} effect sizes (Vargha and Delaney, 2000). Here $A_{12} < 0.5$ means the first arm reaches a lower (better) objective, and effects are called negligible below 0.56, small below 0.64, medium below 0.71 and large otherwise (with the symmetric thresholds below 0.5).

Across domains, arms are ranked by median and compared with a Friedman test (Friedman, 1937).

Because f_1 and f_3 are measured in CPU time, two runs with the same seed are not bit-identical. In FlowShop, Rule A excludes nothing, so SP-CF-AILTA-XAI and its base are the same algorithm there. The difference between those two arms (Table 5) gives a direct estimate of run-to-run noise.

3.8 Computational constraints

All experiments ran on a single machine with 12 CPU cores. The first experiment runner started each run separately through Gradle and executed the runs one after another. Starting the build tool and a new Java virtual machine for every run cost about four minutes per run, which made the 600-run matrix impractical.

The runner was therefore replaced with a concurrent one. Six worker processes ran in parallel, and each worker kept one Java virtual machine alive across many runs, so the start-up cost was paid once per worker rather than once per run. The runner also skipped runs that had already finished, so the matrix could be resumed after an interruption.

Running concurrently increased the number of runs completed per hour, but it cannot shorten an individual run, and PersonnelScheduling runs were long. Each 5,000-application PersonnelScheduling run took about 25 minutes. Our earlier profiling showed that 94.8% of PersonnelScheduling time is spent inside HyFlex's own local search heuristics (a mean of 484 ms per application), and only 1.4% in our instrumentation, so no change to our code could make these runs faster. Multi-point runs are also slower than single-point runs for the same budget: in our earlier 100,000-iteration runs, MP-CF took 713 s on BinPacking against about 200 s for each single-point technique, and 28,042 s on PersonnelScheduling against 20,572 s to 22,398 s.

The single-point arms were completed first, so the multi-point arms had the least time remaining. When the time available for the project ran out, 220 multi-point runs had not been completed: 40 of the 60 MP-CF-XAI runs, and every MP-CF-XAI-LODO, MP-CF-RAND and MP-CF-STAT run. This is why the multi-point results in Section 4.5 are preliminary.

4 Results

4.1 RQ1: Which heuristics are effective, and when?

All 180 effectiveness surrogates pass the fidelity gate, with ROC-AUC between 0.78 (VRP) and 0.99 (PersonnelScheduling) and SHAP ranking stability between $\tau = 0.49$ and $\tau = 0.85$. This includes the SP-SR-LA surrogates, whose acceptance surrogates in Assessment 5 had a mean ROC-AUC of only 0.52. The change confirms the Assessment 5 diagnosis: what those surrogates lacked was the identity of the current heuristic, not structure in the data.

Domain	Effective types (positive signed SHAP)	Ineffective types (negative signed SHAP)
SAT	Local search, mutation	Crossover, ruin-and-recreate
BinPacking	Ruin-and-recreate, local search	Crossover, mutation
FlowShop	Local search, ruin-and-recreate, mutation	None meet all exclusion conditions
TSP	Ruin-and-recreate, local search	Crossover, mutation
VRP	Mutation, local search, ruin-and-recreate	Crossover
PersonnelScheduling	Local search, mutation, crossover	Ruin-and-recreate

Table 3: Heuristic type effectiveness under random selection (SP-SR-LA, instance 0, ten seeds).

Table 3 summarises the effectiveness map. Three patterns emerge. Local search is the only type with a positive contribution in all six domains. Crossover is never effective in single-point search except in PersonnelScheduling, which is consistent with the Assessment 5 observation that a single-point hyper-heuristic crosses the incumbent with itself, so in BinPacking every crossover application left the objective unchanged. Mutation and ruin-and-recreate are domain-dependent: mutation is effective in SAT, VRP and PersonnelScheduling but not in BinPacking or TSP, while ruin-and-recreate is effective in four domains but harmful in SAT and PersonnelScheduling.

Domain	XAI exclusions by phase	XAI-LODO exclusions by phase
SAT	Crossover, ruin-and-recreate (early, middle)	Crossover (all phases)
BinPacking	Crossover, mutation (all phases)	Crossover (early, middle)
FlowShop	None	Crossover (all phases)
TSP	Crossover, mutation (early, middle); crossover (late)	Crossover (early, middle)
VRP	Crossover (all phases)	Crossover (early, middle)
PersonnelScheduling	Ruin-and-recreate (all phases)	Crossover (all phases)

Table 4: Selection masks produced by Rule A. Leave-one-domain-out (LODO) rules use only the other five domains' maps.

The extracted masks (Table 4) show that effectiveness depends on the phase as well as the domain. In SAT, ruin-and-recreate and crossover are excluded early and in the middle of the search but allowed late. In TSP, mutation is excluded only in the first 70% of the budget. This answers the question that the benchmark study of Yates et al. (2025) poses for the researcher, namely which heuristics are effective at which points, and turns the answer into a form the hyper-heuristic can act on.

4.2 RQ2: Do XAI-derived changes improve the hyper-heuristics?

Domain	Median base	Median XAI	A_{12}	Effect	p_{Holm}	Significant
SAT	23.0	22.0	0.275	Large	0.273	No
BinPacking	0.0170	0.0163	0.110	Large	0.020	**Yes, XAI better**
FlowShop†	6279.5	6283.0	0.610	Small	0.580	No
TSP	117,602	117,336	0.540	Negligible	1.000	No
VRP	20,677	20,686	0.570	Small	1.000	No
PersonnelScheduling	2,312	3,028	0.880	Large	0.109	No

Table 5: SP-CF-AILTA-XAI against unmodified SP-CF-AILTA on held-out instance 1, ten paired seeds. $A_{12} < 0.5$ favours XAI. †Rule A excludes nothing in FlowShop, so the two arms are the same algorithm and the difference reflects run-to-run noise.

Table 5 shows the single-point result. On BinPacking the XAI-derived mask significantly improves SP-CF-AILTA with a large effect ($A_{12} = 0.11$, $p = 0.020$), lowering the median objective by about 4%. This is the first improvement in the project that can be traced directly to a SHAP and LIME finding. SAT moves in the same direction with a large effect size, but the difference is not significant with ten seeds. TSP and VRP are unchanged. PersonnelScheduling gets worse with a large effect ($A_{12} = 0.88$) that does not survive the Holm correction; Section 6 explains why. The multi-point arm is reported separately in Section 4.5 because it is incomplete.

Domain	Base	XAI	XAI-LODO	RAND	STAT
SAT	4.0	2.5	2.5	5.0	1.0
BinPacking	3.0	1.0	2.0	5.0	4.0
FlowShop	3.0	4.0	1.0	2.0	5.0
TSP	4.0	3.0	5.0	2.0	1.0
VRP	2.0	4.0	1.0	3.0	5.0
PersonnelScheduling	1.0	5.0	3.0	4.0	2.0
Mean rank	2.83	3.25	**2.42**	3.50	3.00

Table 6: Ranks by median final objective for the five SP-CF-AILTA arms on instance 1 (1 = best). Friedman test across the six domains: $p = 0.80$.

4.3 RQ3: Do the changes generalise to unseen domains?

The leave-one-domain-out rules reduce, by majority vote, to excluding crossover in single-point search, because that is the only exclusion the other five domains agree on. Table 6 shows that this transferred rule has the best mean rank of the five arms (2.42), ahead of the base technique (2.83) and the in-domain rules (3.25). It is significantly better than the in-domain XAI rules on FlowShop ($A_{12} = 0.11$, $p = 0.049$), a domain whose own map excluded nothing, and it avoids the PersonnelScheduling loss. In-domain XAI is in turn significantly better than XAI-LODO on BinPacking ($A_{12} = 0.84$ for LODO against XAI, $p = 0.010$), where the extra mutation exclusion matters. The Friedman test does not detect an overall difference between the five arms ($p = 0.80$), which is unsurprising with six domains as blocks.

4.4 RQ4: Did the improvement happen for the reason the explanation predicted?

Controls. On BinPacking, XAI beats both the random mask and the statistics-only mask on every seed (Table 7). Removing the same number of type-class and phase cells at random does not reproduce the gain, so the gain depends on *which* heuristics are removed, not on shrinking the heuristic set. Removing them by mean log return alone also does not reproduce it, so the surrogate-based selection, which combines a consistent SHAP sign with a model-free confirmation, picks out something the descriptive statistic misses. On PersonnelScheduling the statistics-only mask beats XAI, and both rank below the base technique (Table 6), so neither mask helps there.

Domain	Comparison	A_{12}	p_{Holm}	Better arm
BinPacking	XAI vs RAND	0.000	0.010	XAI
BinPacking	XAI vs STAT	0.000	0.010	XAI
PersonnelScheduling	XAI vs STAT	0.850	0.010	STAT

Table 7: Significant comparisons between SP-CF-AILTA-XAI and the control masks. No other control comparison is significant after Holm correction.

Behaviour. The explanation predicts that the mask should move applications away from types that rarely improve, raising the improvement rate per application and reducing wasted applications. Figure 2 shows that this is what happens on BinPacking. The improvement rate rises by 62% early, 65% in the middle and 100% late, and the share of wasted applications falls from 23% to 0.1% early and from 24% to 0.3% in the middle. SAT shows a smaller rise of 28% in the early and middle phases, consistent with its non-significant performance trend. On PersonnelScheduling the change goes the wrong way: the improvement rate falls by 44% early and to zero in the middle, which matches the loss of performance.

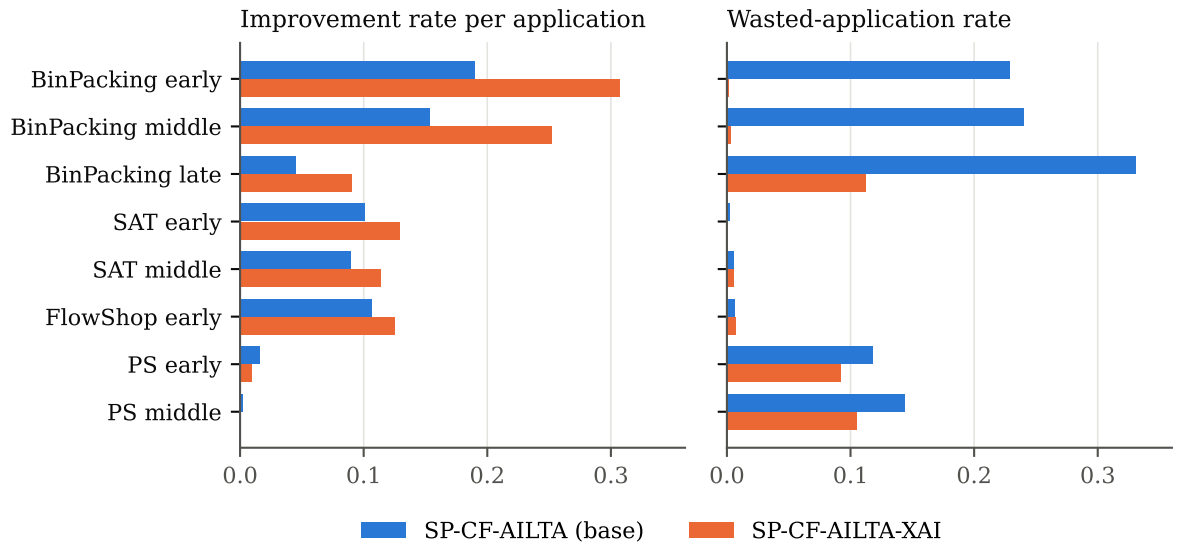


Figure 2: Behaviour of SP-CF-AILTA and SP-CF-AILTA-XAI on instance 1, by domain and phase (mean over ten seeds). Wasted applications are those that leave the objective unchanged or are non-improving and rejected. PS is PersonnelScheduling.

Re-explanation. Sixty effectiveness surrogates retrained on the SP-CF-AILTA-XAI traces all pass the fidelity gate (ROC-AUC 0.73 to 0.99). Excluded types have a SHAP contribution of zero by construction, because they are never applied. The SHAP of the remaining types also shifts: on BinPacking the contribution of local search falls from +1.23 to +0.03 once crossover and mutation are gone. This is expected, because SHAP measures a type's contribution relative to the other types in the data, so once the no-op types are removed, local search no longer stands out against them. Comparing SHAP across surrogates trained on different heuristic distributions is therefore not a reliable test of mechanism, and the behavioural metrics above are the stronger evidence.

4.5 Multi-point search

Domain	MP-CF (10 seeds)	MP-CF-XAI (2 to 4 seeds)	Lower median
SAT	48.0	47.0	XAI
BinPacking	0.0081	0.0118	Base
FlowShop	6313.0	6323.0	Base
TSP	114,190	115,683	Base
VRP	21,174	22,703	Base
PersonnelScheduling	2,280	2,180	XAI

Table 8: Preliminary multi-point median final objectives on instance 1. Too few MP-CF-XAI seeds completed for paired tests, and the LODO, RAND and STAT arms did not run.

Only 80 of the 300 multi-point runs completed (Table 8), for the reasons given in Section 3.8, so no statistical claim is made for MP-CF. The preliminary medians favour the base technique in four of six domains, including BinPacking, where the same rule helped single-point search. One explanation fits both observations. The rules were extracted from single-point traces, where crossover combines the incumbent with itself and so cannot improve it. In MP-CF, crossover has a genuine second parent from the population, so excluding it may remove a useful operator. If this is correct, the explanation is valid for the search topology it was learned from but not for the other, and multi-point rules should be extracted from multi-point traces.

5 Answers to the Research Questions

Table 9 gives the short answer to each research question with the evidence it rests on, and the paragraphs that follow state each answer in full.

Question	Short answer	Main evidence
RQ1 Understand	Yes	180 of 180 surrogates pass the fidelity gate (ROC-AUC 0.78 to 0.99); stable across seeds (τ 0.49 to 0.85); clear, phase-dependent effectiveness patterns (Tables 3 and 4).
RQ2 Improve	Yes, in one domain (single-point only)	BinPacking significantly better ($A_{12} = 0.11$, $p = 0.020$); SAT better but not significant; PersonnelScheduling worse but not significant; multi-point untested (Tables 5 and 8).
RQ3 Generalise	Partly	The transferred crossover rule has the best mean rank (2.42) and never causes a significant loss, but no significant gain on a held-out domain; Friedman $p = 0.80$ (Table 6).
RQ4 Mechanism	Yes on BinPacking; no on PersonnelScheduling	BinPacking: improvement rate up 62% to 100%, wasted applications from 23% to 0.1%, beats random and statistics-only masks on every seed (Table 7, Figure 2).

Table 9: Direct answers to the four research questions.

RQ1: Which heuristic types, in which search states and phases, produce improving moves in each domain? Local search produces improving moves in every domain and phase. Crossover produces almost none in single-point search in every domain except PersonnelScheduling. Mutation and ruin-and-recreate depend on the domain: mutation helps in SAT, VRP and PersonnelScheduling but not in BinPacking or TSP, and ruin-and-recreate helps in four domains but not in SAT or PersonnelScheduling. Several effects also depend on the phase of the search: in SAT, ruin-and-recreate and crossover are ineffective early and in the middle but not late, and in TSP mutation is ineffective only in the first 70% of the budget. These explanations can be trusted because all 180 surrogates pass the fidelity gate and their feature rankings are moderately to strongly stable across seeds.

RQ2: Do changes derived from the RQ1 explanations significantly improve SP-CF-AILTA and MP-CF? For the single-point hyper-heuristic, yes, but only in one of six domains. The XAI-derived mask significantly improves SP-CF-AILTA on BinPacking with a large effect ($A_{12} = 0.11$, $p = 0.020$, median objective about 4% lower). SAT improves with a large effect size that is not significant with ten seeds, TSP, VRP and FlowShop are unchanged, and PersonnelScheduling gets worse with a large effect that is not significant after correction. For the multi-point hyper-heuristic the question cannot be answered, because too few MP-CF-XAI runs completed for a paired test, and the preliminary medians favour the unmodified MP-CF in four of six domains.

RQ3: Do the changes still help on a domain whose traces were not used to derive them? Partly. When each domain is held out, the remaining five domains agree only that crossover should be excluded in single-point search. That transferred rule has the best mean rank of all five arms (2.42 against 2.83 for the base technique), significantly beats the in-domain rule on FlowShop and avoids the PersonnelScheduling loss, but it is not significantly better than the base technique on any held-out domain, and the Friedman test finds no overall difference ($p = 0.80$). The explanation therefore generalises across domains in the sense that its shared part never hurts, but the gains come from domain-specific detail, which also carries the risk.

RQ4: Did the improved variants change behaviour as the explanations predicted, and do they beat equally sized changes not chosen by XAI? On BinPacking, yes on both counts. The explanation predicted that removing crossover and mutation would move applications to types that improve the solution, and the improvement rate per application rose by 62% to 100% while wasted applications fell from 23% to 0.1% early in the search. The XAI mask also beat a random mask and a statistics-only mask of the same size on every seed ($A_{12} = 0.00$), so the gain comes from which heuristics the explanation identified, not from removing heuristics in general. On PersonnelScheduling the prediction failed: removing ruin-and-recreate lowered the improvement rate instead of raising it, and the statistics-only mask did significantly better than the XAI mask.

The overarching question. Taking the four answers together, the overarching question is answered as follows.

1. **Understand: yes, for both search types.** For single-point search, the effectiveness explanations show which heuristic types improve the solution in which domain and phase (RQ1). For multi-point search, the acceptance explanations from our earlier work show that MP-CF's replacement decisions are driven by population diversity and the rank of the replacement target, and effectiveness surrogates were also trained on the MP-CF traces.
2. **Improve: yes, for single-point search.** The XAI-derived mask significantly improves SP-CF-AILTA on BinPacking (RQ2), the improvement happens for the reason the explanation gave, and masks of the same size chosen at random or by descriptive statistics do not reproduce it (RQ4).
3. **Single-point and multi-point: only partly.** Understanding is achieved for both topologies, but improvement is demonstrated only for single-point search. The multi-point improvement step could not be completed on the available hardware (Section 3.8), and the preliminary runs suggest that rules learned from single-point search do not carry over to multi-point search.
4. **Across the CHeSC domains: yes, with a caveat.** All six CHeSC HyFlex domains are used, and the leave-one-domain-out test shows that the part of the explanation shared by all domains transfers to unseen domains without harm (RQ3). The experimental protocol differs from the competition's, so the results are not comparable to published CHeSC rankings.

In short, XAI was used both to understand and to improve a cross-domain selection hyper-heuristic, which is the central aim of the project. The one part not achieved is improving multi-point search.

6 Discussion

What the loop adds. Assessment 5 could say what the acceptance criterion reacted to. This study uses XAI to find a specific inefficiency, remove it, and show with paired tests, controls and behavioural data that removing it improves performance for the predicted reason. That is the "understand and improve" requirement of the brief, demonstrated in full for one domain and single-point search.

Why the Choice Function cannot fix this on its own. If crossover never improves the incumbent in single-point search, one would expect the Choice Function to learn to avoid it. The f_3 term prevents this. Every non-improving step lowers ϕ, which increases the weight on f_3, and f_3 rewards heuristics that have not been used recently. A heuristic that never helps is therefore revisited every time the search stagnates. The 23% to 33% wasted-application rate of the base

technique on BinPacking is consistent with this, and the mask removes the problem without changing the Choice Function itself. This is a case of XAI revealing a behaviour of a well-known selection method that its own adaptation cannot correct.

The transfer gap between random and adaptive selection. Effectiveness learned under random selection is the marginal value of a heuristic when it is applied in an arbitrary state. Under the Choice Function, heuristics are applied in particular states and in particular sequences. In PersonnelScheduling, ruin-and-recreate rarely improves the incumbent directly under random selection, so the rule removed it, but under the Choice Function it appears to act as a diversification step that makes later local search productive: removing it reduced the improvement rate to zero in the middle phase. Similarly, in SAT the Choice Function already selected the excluded types rarely, so the mask changed little. The unconfounded dataset gives an unbiased estimate of a heuristic's direct effect, but not of its role in a sequence, and the f_2 pair term in the Choice Function exists precisely because such sequential effects matter. Rules derived from the target technique's own traces, or from sequence-level features, would address this.

Relation to the benchmark study. Yates et al. (2025) use XAI so that a researcher can see which heuristics were effective at which points of a search. RQ1 reproduces that kind of insight for the HyFlex domains. The loop goes one step further by turning that insight into a rule, testing the rule on held-out data against controls, and checking whether the hyper-heuristic's behaviour changed in the predicted way.

7 Limitations

1. **Multi-point search is incomplete.** Only 80 of 300 multi-point runs finished on the available hardware, despite running six workers concurrently (Section 3.8), so the brief's requirement to examine both single-point and multi-point search is met for the explanation step but not for the improvement step.
2. **Single held-out instance.** Variants were tested on instance 1 only, so the results may not hold on other instances of the same domain.
3. **Rule B not implemented.** The LIME-derived reheating and diversity triggers were specified but not built, so only one kind of XAI-derived change was tested.
4. **Low statistical power across domains.** With six domains as blocks, the Friedman test can detect only large differences, so the non-significant result does not show that the arms are equivalent.
5. **Transfer assumption.** Rules learned under random single-point selection were applied to adaptive selection and to multi-point search, and PersonnelScheduling and the preliminary MP-CF results show that this transfer is not always valid.
6. **Crossover in single-point search.** Part of the crossover finding reflects how single-point search applies crossover (to the incumbent and itself) rather than a property of the domains.
7. **Run-to-run noise.** Time-based Choice Function terms make same-seed runs non-identical. The FlowShop comparison between identical algorithms ($A_{12} = 0.61$, not significant) indicates the size of this noise.
8. **Budgets.** PersonnelScheduling has only 5,000 applications, and BinPacking and FlowShop reached 95% of their improvement close to the 100,000-application pilot cap, so those runs may not be fully converged.

9. **Not comparable to CHeSC.** The domains are the CHeSC HyFlex domains, but the protocol (application budgets, one instance, ten seeds) differs from the competition's timed runs on competition instances, so results are not comparable to published CHeSC rankings.

8 Conclusion

The brief asks for XAI to be used to understand and improve selection hyper-heuristics. This study changes what the XAI explains, from acceptance to heuristic effectiveness, and uses the Simple Random baseline as an unconfounded source of evidence. The resulting explanations are faithful and stable, and they show which heuristic types improve the solution in which domains and phases (RQ1). Turned into a selection mask by a fixed procedure, they significantly improve the single-point Choice Function hyper-heuristic on BinPacking with a large effect (RQ2). The one rule all domains agree on transfers to unseen domains without harm and gives the best average rank, though not a significant gain (RQ3). On BinPacking, the improvement happened for the reason the explanation gave and could not be reproduced by random or purely statistical masks (RQ4). The study also exposes the main limitation of this approach: explanations learned under random selection measure a heuristic's direct effect, not its role in a sequence, and can mislead when that role matters, as in PersonnelScheduling.

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